



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE
NAME

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CENTRE
NUMBER

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INDEX
NUMBER

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CLASS

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H2 BIOLOGY

9648/02

Paper 2 Core Paper

14 September 2011

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Centre number, index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question. Circle the question number in the table on the right.

At the end of the examination, fasten all answer paper securely.

Hand in the **Question Paper** and answers to **Section B** separately.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	/ 80
1	/ 8
2	/ 14
3	/ 14
4	/ 11
5	/ 12
6	/ 11
7	/ 10
Section B	
8 or 9*	/ 20
Total	/ 100

This Question Paper consists of **21** printed pages.

Section A

Answer **all** the questions in this section.

1. Lipid vesicles (i.e. spherical lipid bilayers) that contain $\text{Na}^+\text{-K}^+$ pumps as the sole membrane protein were prepared. For the sake of simplicity, assume that each pump transports one Na^+ one way and one K^+ the other way in each pumping cycle as shown in **Fig. 1.1**. Each $\text{Na}^+\text{-K}^+$ pump is oriented so that the portion of the molecule that normally faces the cytosol faces the outside of the vesicle.

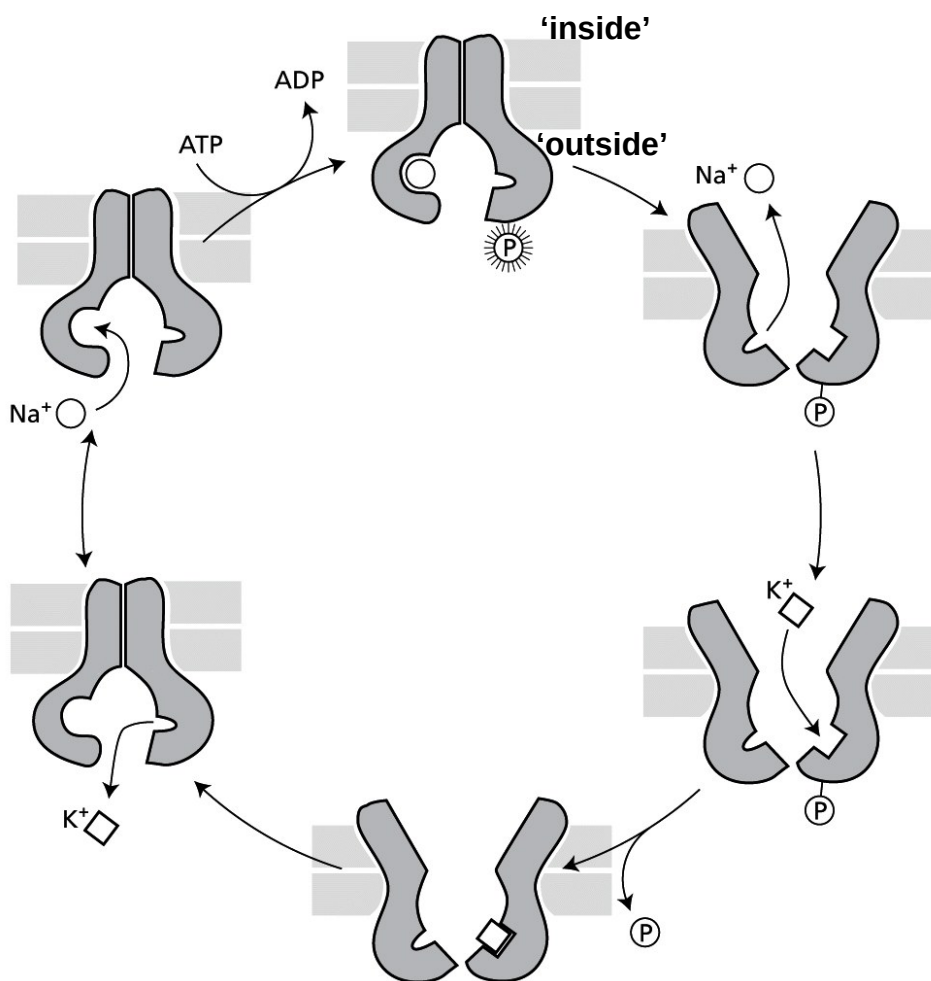


Fig. 1.1

With reference to **Fig. 1.1**, explain what would happen under each of the following conditions.

- (a) The solution inside and outside the vesicles contains both Na^+ and K^+ ions but no ATP.

[1]

1. (b) The solution inside the vesicles contains both Na^+ and K^+ ions; the solution outside contains both ions, as well as ATP. [2]

- (c) The solution inside contains Na^+ ; the solution outside contains Na^+ and ATP. [3]

- (d) Describe how the Na^+ - K^+ pumps are responsible for maintenance of the resting membrane potential in a motor neurone. [2]

[Total: 8]

2.

PART I

In muscle tissue, the conversion of glucose to lactic acid during anaerobic respiration can be represented using a four-stage process as shown in **Fig. 2.1**.

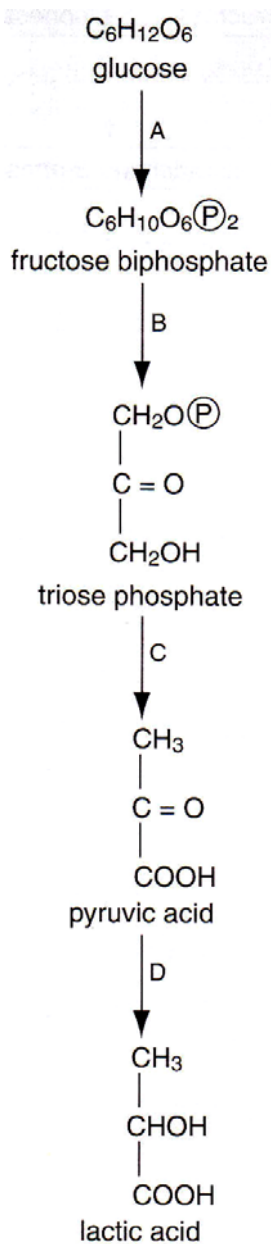


Fig. 2.1

(a) Describe stages **A**, **B**, **C** and **D**.

[4]

A:

B:

C:

D:

2. (b) Is there a net change in the concentrations of NAD^+ and NADPH/H^+ in the whole process? Explain your answer. [2]

- (c) If one glucose molecule is converted to two lactic acid molecules, what is the net synthesis of ATP? Explain your answer. [2]

- (d) What is the metabolic significance to the muscle of this net synthesis of ATP? [1]

2.

PART II

Phosphofructokinase (PFK) has been isolated from the hyperthermophilic archaeon *Pyrococcus furiosus*. It was subjected to standard analysis to determine its basic catalytic properties. The processes under study were of the form

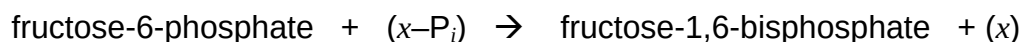


Fig. 2.2 demonstrates the increase in PFK activity with increasing ADP (i.e. $x\text{-P}_i$) concentration. The increase in PFK activity is determined by the increase in fructose-1,6-bisphosphate.

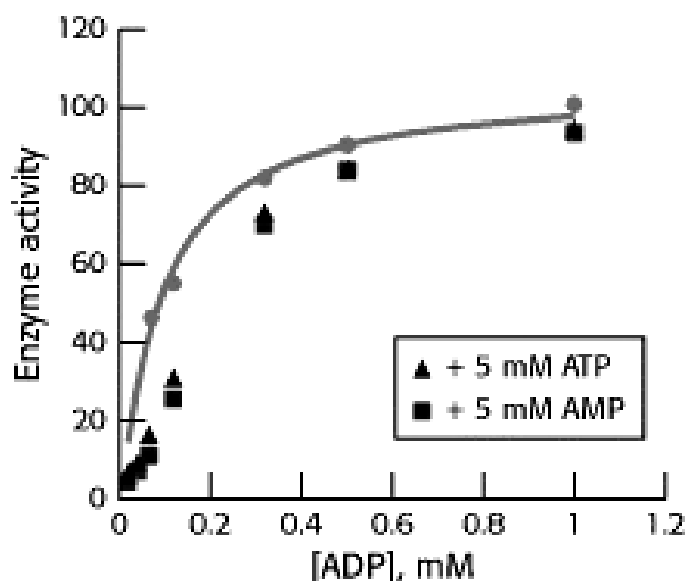


Fig. 2.2

- (e) With reference to the information given, explain how *P. furiosus* PFK differs in action from normal PFK. [2]

2. (f) With reference to **Fig. 2.2**, explain the effect(s) that AMP and ATP have on the reaction with ADP. [3]

[Total: 14]

3.

PART I

Fig. 3.1 is a diagrammatic representation of the process of transcription.

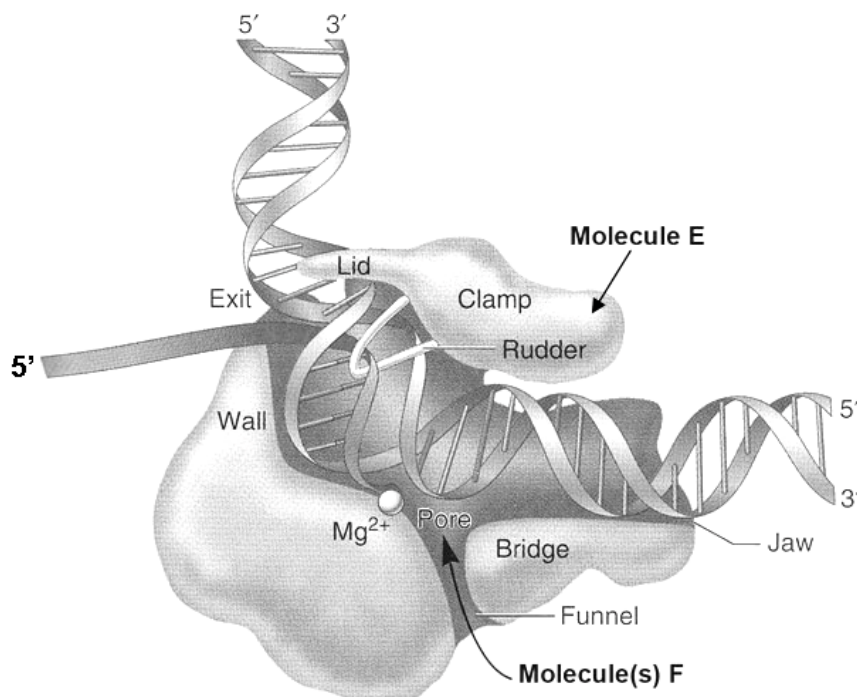


Fig. 3.1

- (a) Describe how the structure of molecule **E** is adapted to its role in transcription. [2]

- (b) Describe two ways by which the structure of molecule(s) **F**, which is / are adapted for transcription, differs from that of amino acids, which are adapted for translation. [2]

1.

2.

3. (c) Using evidence from **Fig. 3.1**, explain why the process of transcription is conservative. [2]

PART II

Double-stranded DNA of the human insulin gene is denatured and allowed to hybridise with mature mRNA isolated from pancreatic cells. Electron microscopy revealed heteroduplex molecules as shown in **Fig. 3.2**.

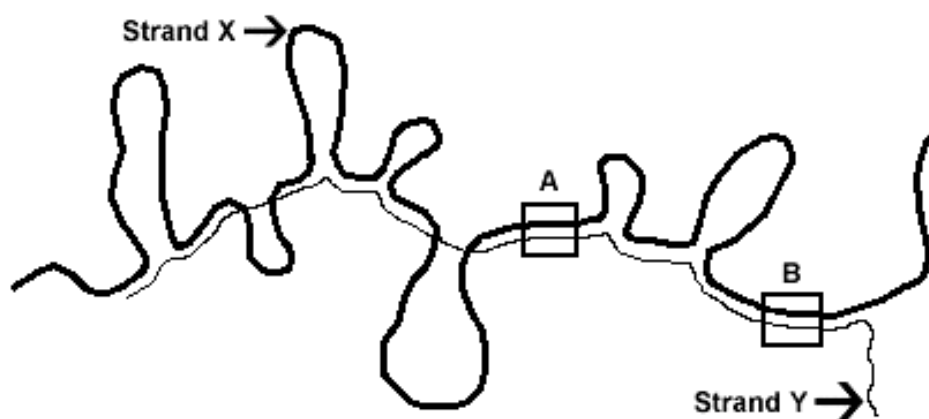


Fig. 3.2

- (d) (i) State the identity of strand X. [1]

- (ii) Using evidence from **Fig. 3.2**, explain the identity of strand X. [3]

3. (e) Circle the locations of the promoter as well as the termination regions of the insulin gene on **Fig. 3.2**. Label the regions accordingly. [2]

(f) What does the unpaired segment on strand **Y** represent? Give a reason for your answer. [2]

[Total: 14]

4.

PART I

Fig. 4.1 is an electron micrograph showing a virus infecting a host cell.

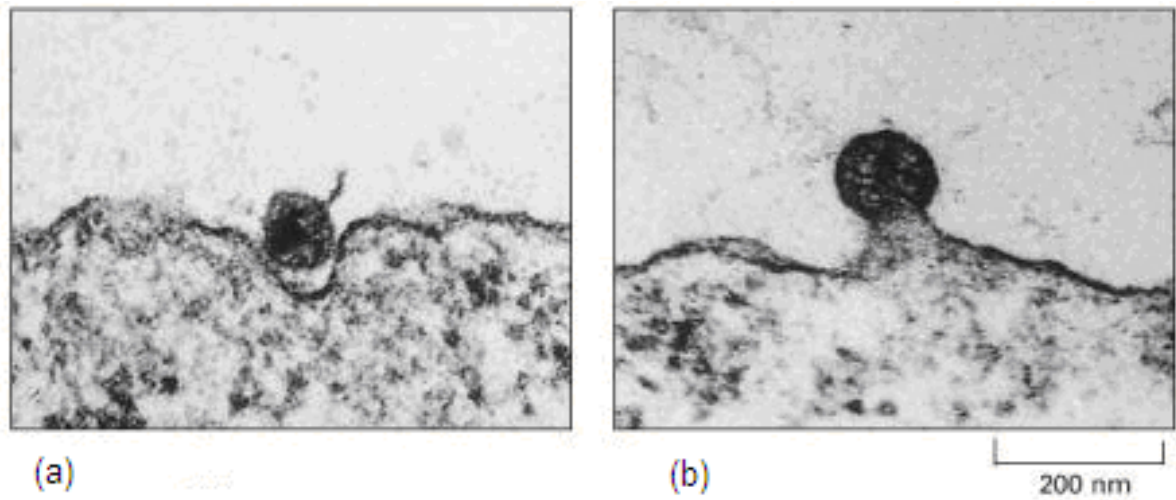


Fig. 4.1

(a) Describe how an enveloped virus such as HIV enters a host cell. [3]

(b) Suggest why the HIV causes damage only to some types of human cell. [1]

4.

PART II

Several studies have already revealed that some CCR5 receptors belonging to the G protein-coupled receptors family might be necessary for the entry of the virus into the cell.

Fig. 4.2 is a 3-dimensional model of the human CCR5 receptor, which is a protein.

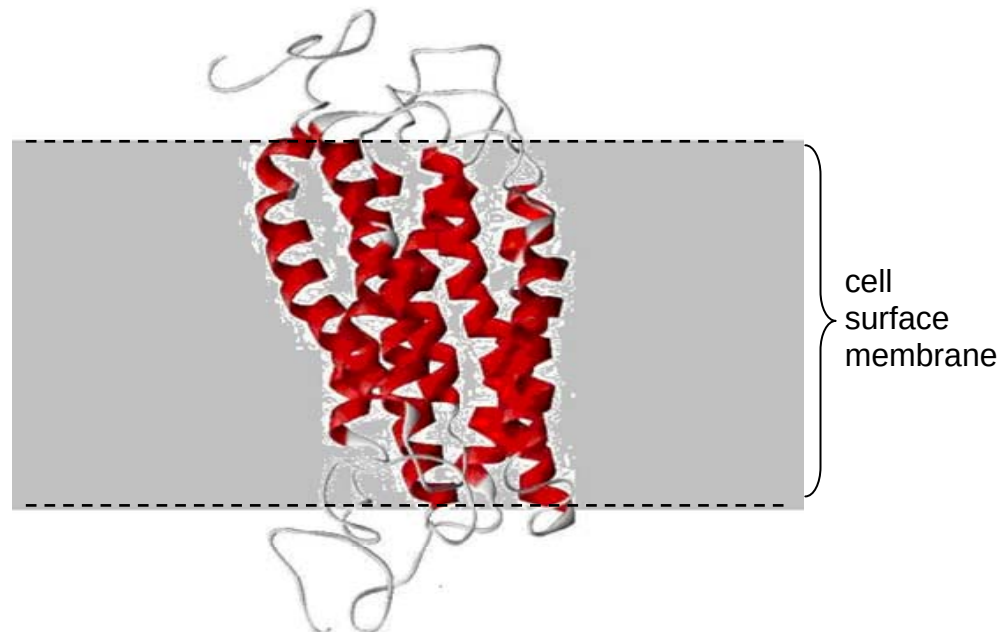


Fig. 4.2

- (c) With reference to **Fig. 4.2**, describe how the polypeptide chain of CCR5 allows it to be maintained in the membrane.

[4]

4. It was observed that a small percentage of high-risk populations showed either resistance or delayed development of the disease. The resistance of these rare individuals to HIV infection has now been explained by the discovery that they are homozygous for an allelic, nonfunctional variant of CCR5 caused by a 32-base-pair deletion from the coding region in the gene that codes for the CCR5 receptor.

(d) Explain **one** possible consequence of how the mutation results in resistance to the disease. [3]

[Total: 11]

5. The colour of the spines on the stems of raspberry plants are controlled by two genes, **A/a** and **B/b**. The genes are on different pairs of chromosomes.

Allele **A** produces a pink anthocyanin pigment in the spines. Allele **B** has no effect by itself, but increases the colour produced by allele **A** to give red spines. Alleles **a** and **b** have no effect on spine colour. In the absence of anthocyanin, the spines are green.

- (a) Suggest how allele **B** may alter the expression of allele **A**. [2]

Plants with the genotypes **AaBb** and **aabb** were cross-pollinated. The resulting seeds were sown and the seedlings grown until their stems developed spines.

- (b) (i) Draw a genetic diagram of this cross to show the ratio of different phenotypes expected in the offspring. [4]

5. (b) (ii) In any genetic cross, the observed results are usually different from the expected results.

Suggest *two* reasons why such a difference may occur, referring only to events after meiosis. [2]

1.

2.

- (iii) Explain what differences in the phenotypic ratio would be expected if genes **A/a** and **B/b** were on the same homologous pair of chromosomes, as shown in **Fig. 5.1**.

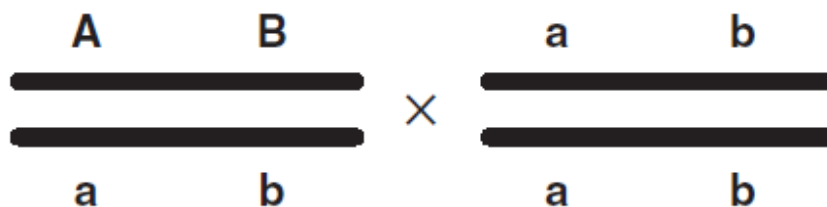


Fig. 5.1

[4]

[Total: 12]

6. In Lake Tanganyika in Africa, there are six species of fish of the genus *Tropheus* and a much larger number of distinctly coloured subspecies of each of the six species. *Tropheus* species are small fish that are confined to isolated rocky habitats around the shores of Lake Tanganyika.

The six species evolved during the primary radiation phase when the lake was first filled, about 1.25 million years ago. They arose from river dwelling ancestors and then filled all available niches in the lake.

Secondary radiations into the many subspecies occurred during the last 200 000 years. Sometime during this period, the water level in the lake fell, resulting in the formation of three separate lake basins. These basins persisted for many thousands of years before the water level rose again.

Fig. 6.1 shows an outline map of the lake and the location of the three temporary basins caused by lowering of lake levels.

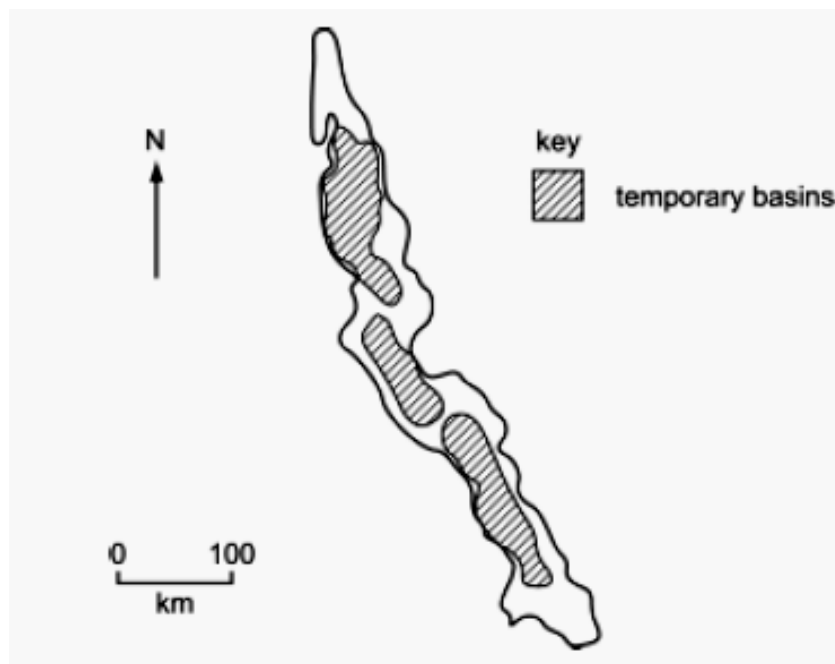


Fig. 6.1

- (a) (i) Explain how natural selection could have caused the evolution of the six closely related species in the primary radiation. [3]

6. (a) (ii) Suggest how the lowering of the water level in the lake to form three separate lake basins could have caused the evolution of so many subspecies. [2]

Recent research has compared DNA sequence of cytochrome b gene of the six different species of *Tropheus* (Fig. 6.2).

<i>T. brichardi</i>	CTTTTAACTATGATAACAGCTTTTGTGGGCTACG
<i>T. kasabae</i>	CTTTTAACTATGATAATAGCTTTTGTGGGCTACG
<i>T. annectens</i>	CTTTTAACTATGATAACAGCTTTTGTGGGCTACG
<i>T. moorii</i>	CTTTTAACTATGATAACAGCTTTTGTGGGTTACG
<i>T. polli</i>	CTTTTAACTATAATAACAGCTTTTGTGGGCTACG
<i>T. duboisi</i>	CTTTTAACCATGATAACGGCTTTTGTAGGCTATG
	* *

Fig. 6.2 shows part of the cytochrome b DNA sequence

A family tree based on differences between the DNA sequences is shown in Fig. 6.3.

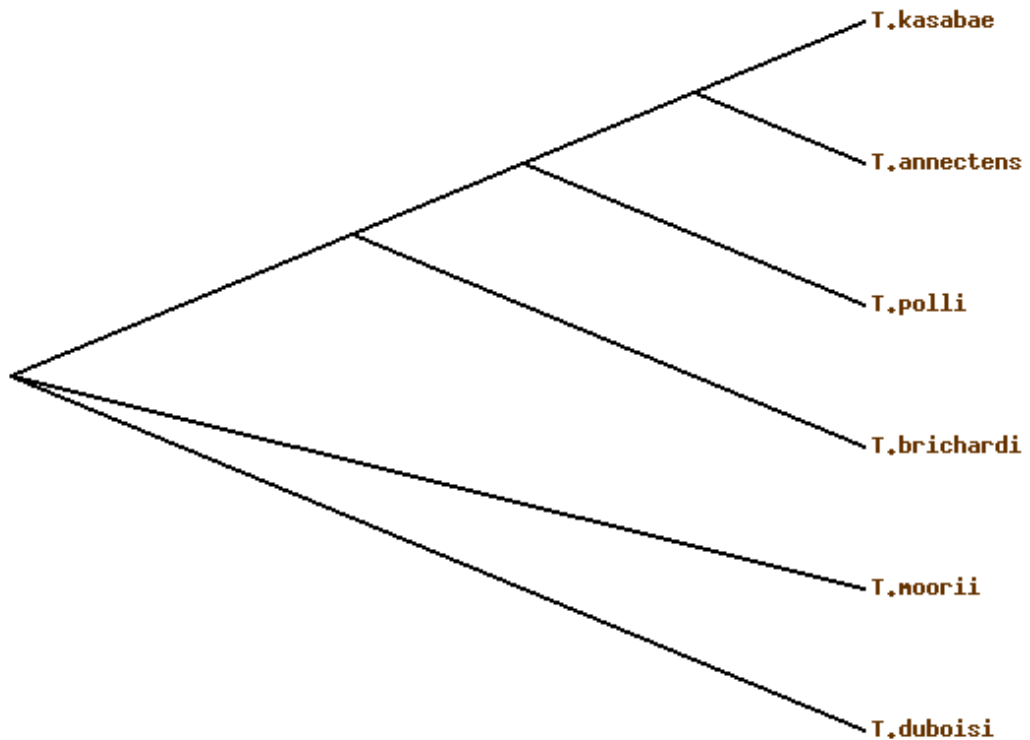


Fig. 6.3

6. (b) With reference to **Fig. 6.2**, explain why different species of *Tropheus* can have similar DNA sequence in their cytochrome b gene. [2]

- (c) With reference to **Fig. 6.2 and 6.3**,

- (i) describe how the time of divergence of the species could be estimated. [2]

- (ii) explain the advantage of the use of DNA sequence over amino acid sequence in determining the phylogeny of *Tropheus* species. [2]

[Total: 11]

7. Diabetes mellitus is often due to the destruction of the β -cells in the islets of Langerhans in the pancreas. **Fig. 7.1** shows the changes in blood glucose concentration over a day in which an untreated diabetic person and a non-diabetic person take the same meals and exercise. Blood glucose was measured in both individuals every hour. At mealtimes, the measurements were made just before the meal.

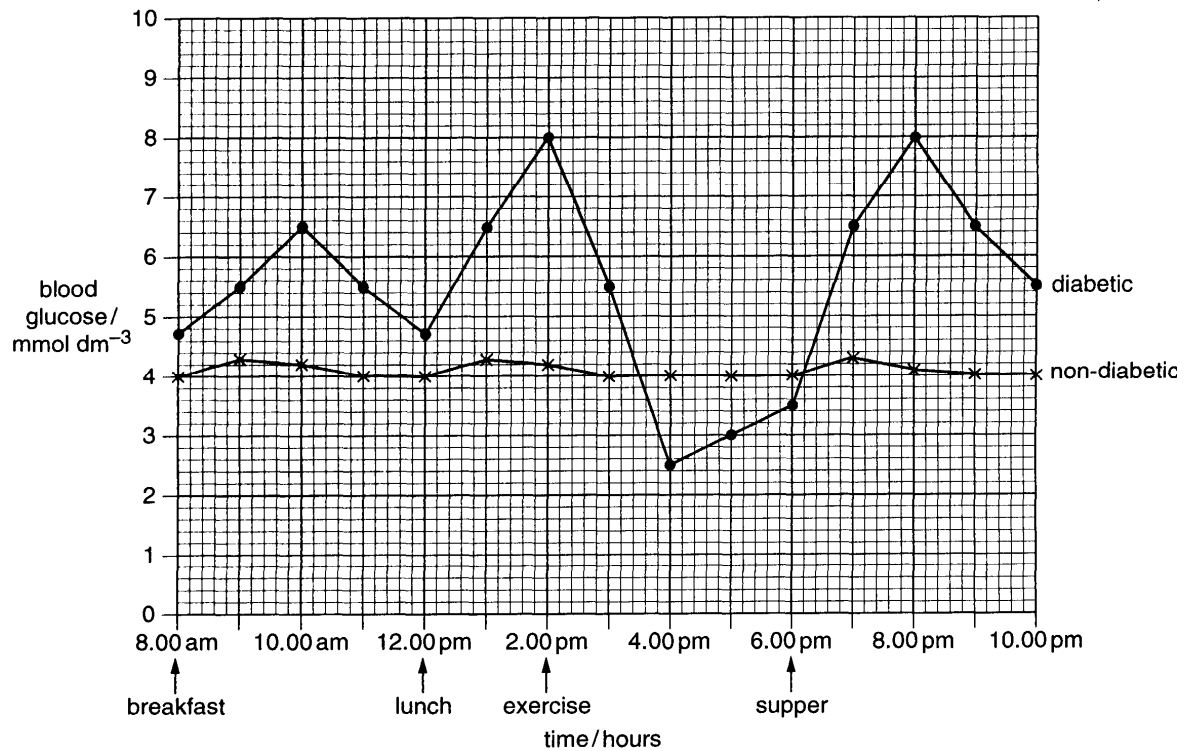


Fig. 7.1

- (a) Account for the differences in blood glucose at 10.00 am.

[3]

7. (b) Explain why, following exercise at 2.00 pm,

(i) the blood glucose in the diabetic person dropped; [2]

(ii) the blood glucose in the non-diabetic person stayed the same. [2]

(c) Suggest a possible cause of the increase in the blood glucose concentration of the diabetic person at 4.00 pm. [1]

(d) Diabetes can be treated by regular injections of a hormone.
State **two** sources of that hormone. [2]

1.

2.

[Total: 10]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL** return is necessary if you have not attempted this section.

8. (a) Describe the behaviour of DNA during the mitotic cell cycle. [8]
- (b) Using named examples, explain how loss of function mutation and gain of function mutation can contribute to cancer. [8]
- (c) Describe the differences between prokaryotic and eukaryotic mature mRNAs. [4]

[Total: 20]

9. (a) Explain how the structure and properties of triglycerides are related to their functions in living organisms. [6]
- (b) Explain the role of membranes in photosynthesis and respiration. [8]
- (c) Compare haemoglobin with insulin in term of structure and function. [6]

[Total: 20]